

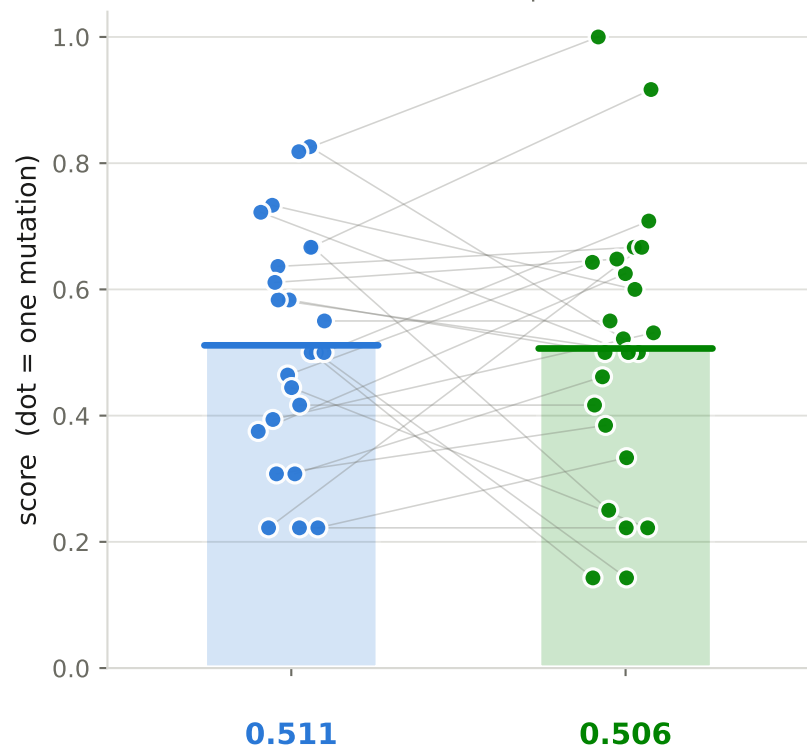
Multimodal vs bulk RNA-seq alone — 24 driver mutations

paired donor-grouped CV-OOF on the single-cell AML atlas · identical samples & folds

■ Bulk RNA-seq alone (no imputation) ■ MATRIX-AML multimodal (8 modalities)

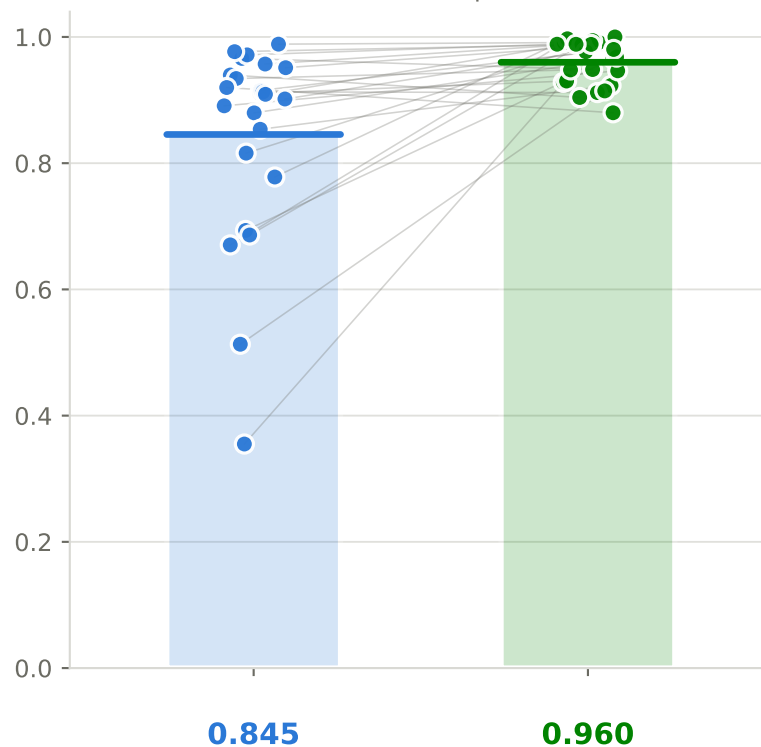
Sensitivity

$\Delta -0.005$ Wilcoxon $p = 0.96$



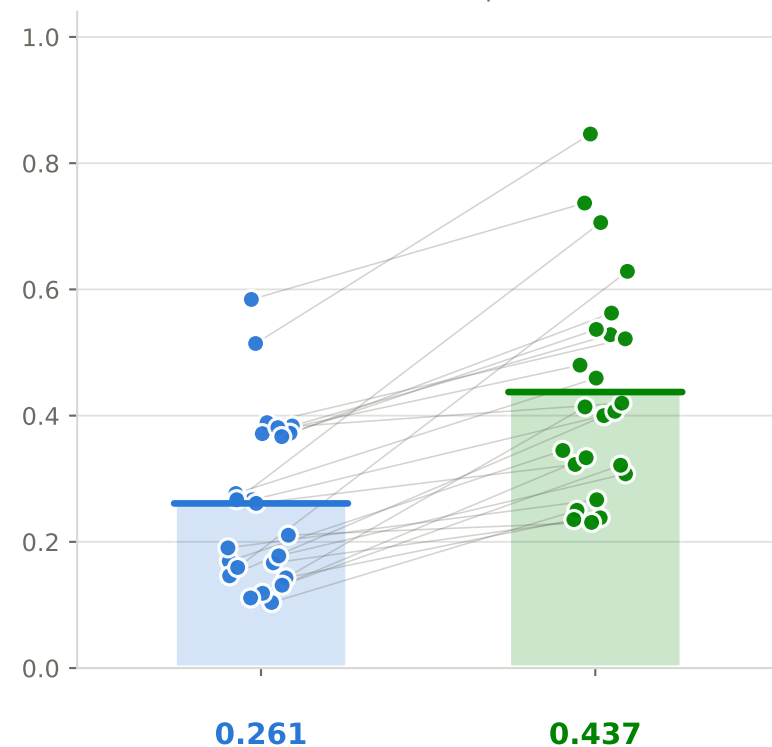
Specificity

$\Delta +0.114$ Wilcoxon $p = 0.00018$



F1

$\Delta +0.176$ Wilcoxon $p = 1.2e-07$



Donor-grouped 3-fold cross-validated out-of-fold predictions; both arms use identical samples, labels and folds.
Operating point = F1-max threshold on the same OOF vector, so absolute values are optimistic — the CONTRAST between arms is the result.